Alicia Freel

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Ph.D. Candidate in Security Informatics | Human-AI Interaction & Responsible AI

EDUCATION

Indiana University Bloomington, IN
Doctor of Philosophy in Security Informatics. Cognate: Cybersecurity;
Minor: Law, Maurer School of Law. *Expected May 2028*
Dissertation: Responsible AI: Design, Reliance, and Accountability in AI-Assisted Decision Making

Indiana University Bloomington, IN
Master of Science in Cybersecurity & Risk Management. *May 2022*
Relevant Coursework: Multidisciplinary degree in Informatics, Law, and Kelley School of Business, Key Projects: Policy analysis for the Department of Defense; legal implications of cybersecurity governance for non-profits

Indiana University Bloomington, IN
Bachelor of Arts in Cognitive Science. Cognate: Computer Science;
Minors: Human-Computer Interaction, Business. *May 2020*

RESEARCH INTERESTS

- Human-AI Interaction, Responsible AI, AI Ethics & Governance, Technology Law & Policy, Cybersecurity, Trust in AI

PUBLICATIONS

Under Review

- **Freel, A.,** Pias, S.B. H., McGraw, E., Sabanovic, S., & Kapadia, A. (2026). Human-First Does Not Mean Influence-Free: How Social Support Shapes Decisions After AI Disagreement. Under review at ACM CHI 2027.

Peer-Reviewed Conference Publications.

- **Freel, A.,** Pias, S. B. H., Sabanovic, S., & Kapadia, A. (2025). How misclassification severity and timing influence user trust in AI image classification: User perceptions of high- and low-stakes contexts. Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT), Athens, Greece.

Workshop Proceedings & Posters

- **Freel, A.,** Pias, S. B. H., Sabanovic, S., & Kapadia, A. (2025). The influence of AI personality on user compliance and ethical behavior. In Proceedings of the STAIG Workshop at ACM CHI 2025, Yokohama, Japan.
- Pias, S. B. H., **Freel, A.,** Huang, R., Williamson, D., Kim, M., & Kapadia, A. Building trust through voice: How vocal tone impacts user perception of attractiveness of voice assistants. Preprint.
- **Freel, A.,** Pias, S. B. H., Sabanovic, S., & Kapadia, A. (2024). Examining trust erosion in human-AI collaboration: Unpacking the impact of severity and timing of misclassification. In Proceedings of the TREW Workshop at ACM CHI 2024, Honolulu, Hawaii.
- Pias, S. B. H., Akter, T., **Freel, A.,** Williamson, D., Trammel, T., & Kapadia, A. (2024). The drawback of insight: Detailed explanations can reduce agreement with explainable artificial intelligence (XAI). In Proceedings of the ACM CHI Workshop, Honolulu, Hawaii.
- Nawshin, S., **Freel, A.,** Kim, M., & Kapadia, A. (2024). Are familiar voices more believable? Deepfake technologies. In Proceedings of the ACM CHI Workshop, Honolulu, Hawaii.

PRESENTATIONS

- Freel, A., Pias, S. B. H., Sabanovic, S., & Kapadia, A. (2025, September). The influence of AI personality on user compliance and ethical behavior. Presented at the Midwest Security Workshop (MSW 10), Bloomington, Indiana.
- Caven, P. J., Raymond, A., & Camp, L. J. (2025, July). Increasing a product's perceived cybersecurity by indicating the country of origin. Presented at HAISA 2025, Mytilene, Greece (presented on behalf of Prof. Jean Camp's lab).
- Freel, A., Pias, S. B. H., Sabanovic, S., & Kapadia, A. (2025, June). How misclassification severity and timing influence user trust in AI image classification: User perceptions of high- and low-stakes contexts. Presented at the ACM Conference on Fairness, Accountability, and Transparency (FAccT 2025), Athens, Greece.
- Freel, A., Pias, S. B. H., Sabanovic, S., & Kapadia, A. (2025, April). The influence of AI personality on user compliance and ethical behavior. Presented at the STAIG Workshop, ACM CHI 2025, Yokohama, Japan.
- Pias, S. B. H., Akter, T., Freel, A., Williamson, D., Trammel, T., & Kapadia, A. (2024, September). Assessing human agreement with artificially intelligent classifiers based on explanation detail. Presented at the Scalable Asymmetric Lifecycle Engagement and Trusted AI Symposium, WestGate Crane NSWC.

- Freel, A., Pias, S. B. H., Sabanovic, S., & Kapadia, A. (2024, May 11). Examining trust erosion in human-AI collaboration: Unpacking the impact of severity and timing of misclassifications. Presented at the TREW Workshop, ACM CHI 2024, Honolulu, Hawaii.
- Freel, A. (2023, March). Trustworthiness of Familiar Voices. Presented at the Indiana Statewide Cybersecurity Summit, Purdue University.
- Pias, S. B. H., Akter, T., Freel, A., Williamson, D., Trammel, T., & Kapadia, A. (2022, April 29). Assessing Human agreement with artificially intelligent classifiers based on explanation detail. Presented at the Trusted AI Symposium, IUPUI.
- Freel, A. (2022, October). Assessing Human agreement with artificially intelligent classifiers based on explanation detail. Presented at the SCALE ESS - Trusted AI Symposium, Indiana University.

RESEARCH EXPERIENCE

Research Assistant February 2021 – Present
Indiana University *Bloomington, IN*

- Investigate how AI interface design, error characteristics, and social influence shape human trust, reliance, and decision-making, with implications for responsible AI and human autonomy.
- Design and conduct controlled human-subjects experiments, including factorial and repeated-measures designs, to isolate the effects of AI misclassification timing, severity, and consensus cues on user behavior.
- Develop and validate experimental stimuli, AI-assisted decision-making scenarios, and survey instruments, conduct pilot studies to assess experimental validity and refine study procedures.
- Conduct quantitative and qualitative analyses of experimental data, including regression modeling, analysis of interaction effects, and thematic coding of participant responses.
- Develop first-author scholarly manuscripts and present research findings at international conferences, connecting empirical results with questions of AI accountability and ethical system design.

Research Intern October 2020 – February 2021
CyberPeace Institute *Geneva, Switzerland (Remote)*

- Researched legal, technical, and policy dimensions of transparency and accountability for malicious actors in cyberspace, domestically and internationally.
- Conducted in-depth analysis of 33 large-scale worldwide cybersecurity attacks, identifying the type of law or norm affected and analyzing technical aspects including attack type, patterns, and attacker and victim profiles.
- Created a taxonomy of common methods for attributing bad actors, informing victims and researchers on what data to collect to support successful attribution.
- Authored a 50-page deliverable to the CEO covering methodology, context, key findings, impacts, conclusions, and future research directions.

Cybersecurity Liaison Intern June 2021 – December 2021
REN-ISAC *Bloomington, IN*

- Engaged with over 600 higher education institution officials, including CISOs, CIOs, and CTOs, to research and understand cybersecurity threats facing domestic and international institutions.
- Analyzed threat intelligence and response data from over 500 private and public institutions to identify trends, best practices, and opportunities to improve cybersecurity posture across higher education.
- Contributed to the development of policy recommendations for improving cybersecurity across academic institutions.

RESEARCH METHODS & TECHNICAL SKILLS

- **Experimental Methods:** Human-subjects experiments, between-subjects and mixed-factorial designs, randomized assignment, stimulus development and validation, survey design, behavioral measurement.
- **Statistical Methods:** Linear and logistic regression, generalized linear mixed-effects models (GLMMs), ANOVA, hypothesis testing, interaction effects, post-hoc comparisons.
- **Qualitative Methods:** Thematic analysis, inductive coding, codebook development, open-ended response analysis, semi-structured interviews.
- **Research Software:** Qualtrics, Prolific, MAXQDA, Delve; RStudio

PROFESSIONAL EXPERIENCE

Vendor Information Protection / Security Engineering Analyst / Risk Engineer Intern May 2021 – Present
Northwestern Mutual *Milwaukee, WI*

- Conducted over 300 vendor contract reviews to identify security provision gaps and assessed contractual obligations under NYDFS 23 NYCRR 500, NIST, ISO 27001, and SOC 2 requirements.
- Led over 50 end-to-end asset risk assessments, each encompassing a technical security engineering assessment, privacy impact assessment, compliance by design and data governance assessment, and issue remediation.

- Review and influence vendor contract language related to data protection, breach notification, audit rights, encryption, incident response, and regulatory cooperation.
- Translate legal and regulatory requirements into operational security expectations in collaboration with Legal, Privacy, and business stakeholders.
- Develop structured evaluation frameworks to support defensible, repeatable risk decisions under regulatory scrutiny.
- Support audit readiness and regulatory examinations by improving documentation quality, traceability, and decision rationale.

Cybersecurity Liaison Intern March 2021 – June 2021
The Mako Group *Carmel, IN*
- Created a ransomware risk assessment audit guide, enhancing client security awareness.
- Improved client security posture by identifying vulnerabilities through vishing assessments in the health industry.

APPLIED RESEARCH & CLINICAL PROJECTS

Clinical Experience February 2021 - May 2021
Growing Places Indy *Indianapolis*, IN
- Organized a cybersecurity best practices and procedures guide using the Ford Foundation Assessment.
- Developed a secure asset management system and prepared an anti-phishing guide.

Clinical Experience January 2021 - May 2021
National Security Agency *Bloomington*, IN
- Developed national strategies for the NSA to partner with public and private sectors to identify malicious actors.
- Identified commercial and private opportunities across firms, software, infrastructure, security devices, and IoT.

Clinical Experience August 2020 - December 2020
Naval Surface Warfare Center (CRANE) *Bloomington*, IN
- Drafted legal contracts and technical solutions to aid in securing network devices in line with NIST 800-53.
- Researched supply chain logistics and methods for implementing better practices to prevent device tampering by foreign states and other bad actors.
- Created a deliverable presented to the Department of Defense.

AWARDS & HONORS

- Luddy Doctoral AI Fellowship Recipient, Indiana University, 2022
- CyberCorps Scholarship for Service, National Science Foundation, 2022

ACTIVITIES & LEADERSHIP

Women's Ultimate Frisbee Club August 2016 – Present
Non-profit Founder, President, Team Captain
- Establish and maintain non-profit status and led the Board of Directors with the VP and executive committee in managing team finances, philanthropy, volunteering, communications, and recruitment.
- Collaborate with teams across the nation to organize practices, tournaments, and travel while leading the team toward its goals and participating in community volunteering events.

Indiana University September 2020 – August 2021
Club Sports Finance Committee Member
- Provided financial analysis and guidance to all university club sports, supporting clubs in improving financial literacy.
- Attended weekly meetings focused on improving and utilizing available resources.